

NATIONAL TECHNICAL RESEARCH ORGANISATION

SIGNAL ANALYSIS & PARAMETER EXTRACTION DOSSIER

CLASSIFICATION: SECRET // AIR-GAPPED INTERNAL // REF: c158a7cbcc7d8103

INTERCEPT METADATA & CHAIN OF CUSTODY

Source Filename : sample_am_dsb.iq
File SHA-256 Digest : c158a7cbcc7d810329ad41834a75d000872b0bf49e7578351cc4dcf1e1b838f9
Payload File Size : 524,288 bytes (512.0 KB)
Analysis Timestamp : 2026-09-18T19:53:00Z
Operator ID : SIGINT-AIRGAP-OPERATOR
AMC Model Engine : amc_resnet18_v1.0 (Deep 1D ResNet-18 + Ensemble)

EXTRACTED RF SIGNAL PARAMETERS

Parameter Description	Estimated Value	Extraction Metric / Method
Sampling Frequency (F_s)	5.0000 MSps	Baseband Nyquist Rate
Center Frequency (F_c)	+0.080 kHz	Spectral Centroid Offset
Estimated Baud Rate	184.17 kBaud	Cyclostationary Wavelet Energy
3-dB Bandwidth	3.66 kHz	Half-Power Frequency Span
Occupied Bandwidth (99%)	2987.06 kHz	Cumulative PSD 99% Energy
Estimated SNR	17.94 dB	Fourth-Order Cumulant (C_{42})
Peak-to-Average Power (PAPR)	7.19 dB	Temporal Envelope Crest Ratio
Carrier Frequency Offset (CFO)	+0.00 Hz	Non-Linear M-th Power Loop
Spectral Flatness (Entropy)	0.0308	Wiener Entropy Index (0.0 – 1.0)

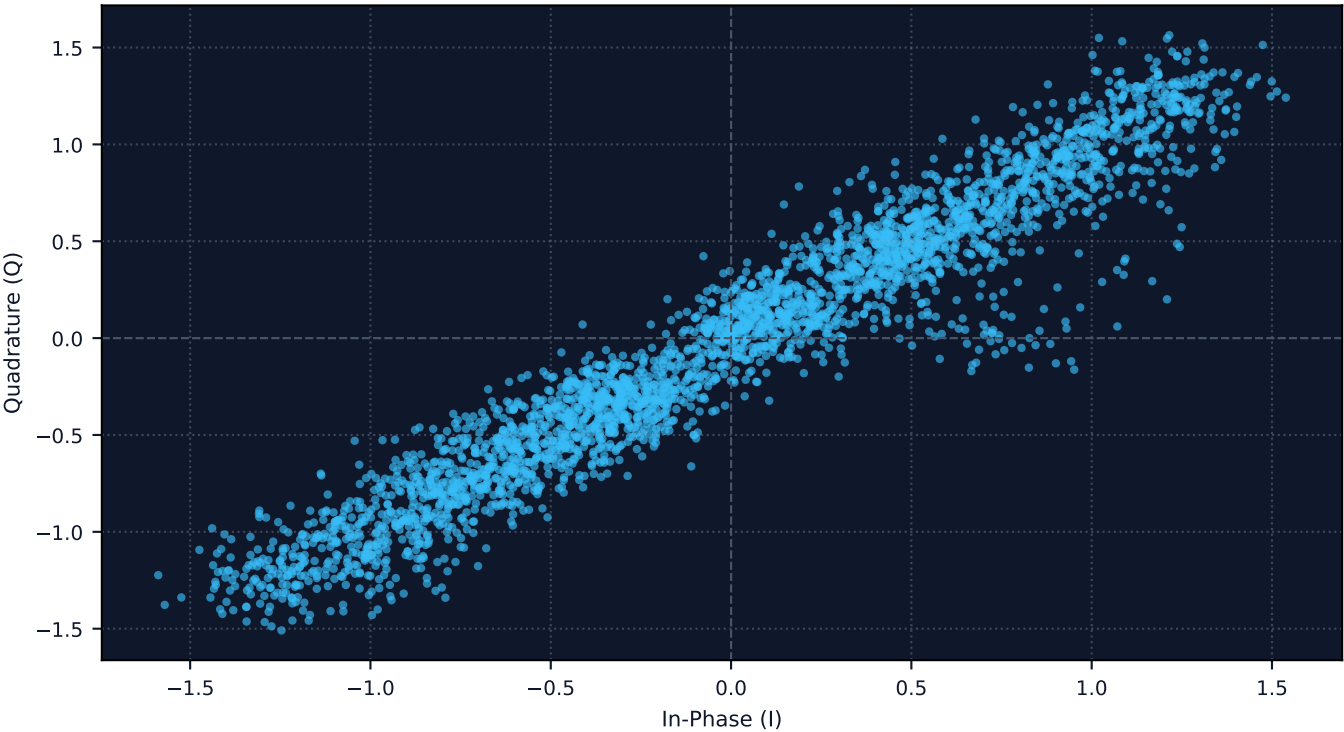
PREDICTED MODULATION : AM-DSB

CLASSIFIER CONFIDENCE : 99.50% (ResNet-18 Softmax Ensemble)

INTELLIGENCE STATUS : CONFIRMED: HIGH CERTAINTY MODULATION CLASSIFICATION

Demodulation Scheme: QPSK | Recovered Bits: 131,072 | Estimated EVM: 10.00% | CRC Validated Frames: 0

Demodulated Baseband Constellation Diagram — [AM-DSB]



SYNCHRONIZED TELEMETRY FRAMES & PROTOCOL DISSECTION

Frame #	Sync Index	Tx ID	Seq No.	Payload	CRC Status	FEC Status
#1	23	0x0000	0	16374 B	CRC MISMATCH	PARTIAL_RECOVER
#2	3771	0x0000	1	15905 B	CRC MISMATCH	PARTIAL_RECOVER
#3	5222	0x0000	2	0 B	CRC MISMATCH	CRC_MISMATCH
#4	5605	0x0000	3	0 B	CRC MISMATCH	CRC_MISMATCH
#5	11219	0x0000	4	14974 B	CRC MISMATCH	PARTIAL_RECOVER

DECODED BITSTREAM HEX / ASCII PREVIEW (FIRST 64 BYTES)

0000 DFDF7FFFFFFFFFFFFFFFFFFFFFFFFF0

[FRAME 1 PAYLOAD]